1. Observe
(Unlabeled Target Data)

Source Reference (CORSET, cached)
Offline; shipped with model



K-means coreset selection



Cached source features and statistics

- Region features (for CKA)
- Distribution statistics (for Fréchet gate)

Target domain (unlabeled)



Unlabeled observations from the target deployment environment

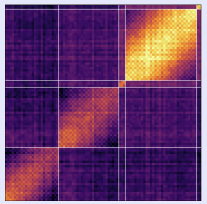
2. Diagnose

Pre-trained VLA (e.g., OpenVLA-OFT)

SigLIP (Vision Encoder) | DINOv2 (Vision Encoder) | Projector (Cross-Modal) | LLaMA (Language Backbone) | Action Head (parallel L1)

Tier-1: Reference-based Comparison (from CORSET)

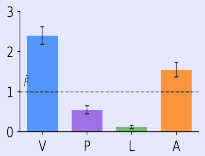
CKA Similarity (Representation Shift)



$$\mathrm{CKA}(r) = \frac{\langle \Sigma_S, \Sigma_T \rangle_F}{\|\Sigma_S\|_F \, \|\Sigma_T\|_F}$$

Representation shift:
1 − CKA(r)

Fisher Information (Parameter Sensitivity)

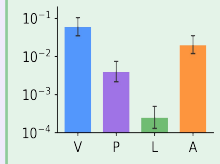


$$F(r) = \frac{1}{|\theta^{(r)}|} \sum_i F_{ii} \Big/ \bar{F}$$

Does this region matter for the output?

Tier-2: Reference-free Signals (unlabeled)

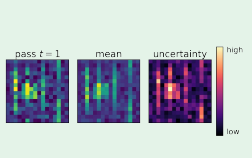
Gradient Norms (Optimization Sensitivity)



$$G(r) = \frac{\|\nabla_{\theta^{(r)}} \mathcal{L}_{\mathrm{surr}}\|_2}{\sqrt{\dim(\theta^{(r)})}}$$

Gradients indicate how much the region needs to change.

MC Dropout Uncertainty (Predictive Uncertainty)



$$\hat{U}(r) = \frac{1}{T} \sum_{t=1}^{T} \|f_r^{(t)} - \bar{f}_r\|_2^2$$

$T = 20$, $p = 0.1$
(Octo / DTP only)

Fréchet Distance Gate (Blend Tier-1 and Tier-2)

Source features (μ_S, Σ_S)

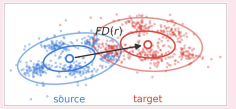


Target features (μ_T, Σ_T)



$$FD(r) = \tfrac{1}{d_r}\left[\|\mu_S - \mu_T\|_2^2 + \mathrm{Tr}(\Sigma_S + \Sigma_T - 2(\Sigma_S\Sigma_T)^{1/2})\right]$$

$$w(r) = \exp\left(-\gamma \cdot FD(r)\right)$$



Smaller domain gap → trust Tier-1 more

Per-Region Adaptation Cost (Final Score)

$$D(r) = w(r)\, r_{\mathrm{rel}}(N)\, \hat{D}_{\mathrm{cmp}}(r) + \left(1 - w(r)\, r_{\mathrm{rel}}(N)\right) D_{\mathrm{fall}}(r)$$

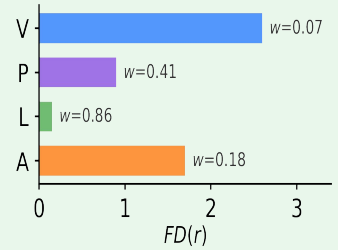
3. Allocate
(Variable-Rank Adapters)

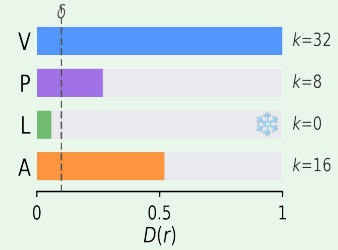
Budget-Aware Rank Allocation

$$k(r) = \begin{cases} 0 & \text{if } D(r) < \delta \\ \max\left(1, \left\lfloor \frac{D(r)}{\sum_{r'} D(r')} \cdot \frac{P_{\mathrm{total}}}{c_1(r)} \right\rfloor\right) & \text{otherwise} \end{cases}$$

$\delta = 0.1$ (freeze threshold); $c_1(r)$ = per-rank param. cost

Example Allocation (per region)





❄ Frozen 🔥 Trainable

Parameter Budget

Total trainable parameters
(e.g., 0.04% – 2% of model)

4. Adapt
(Efficient Fine-Tuning)

Train Only Allocated Adapters

$$\theta = \theta_0 + \Sigma_r B_r A_r$$

All other parameters frozen



Adapted VLA
(real-world deployment)

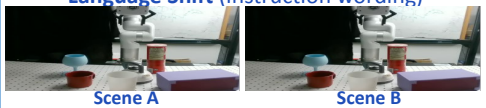
Generalizes to held-out scenes





Real-World Adaptation Results (xArm-7)

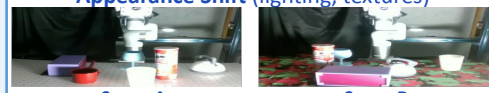
Language Shift (instruction wording)



Scene A "Pick the teapot" 21/30 | Scene B "Pick the kettle." 16/30

0.041%
of full-model parameters
3.1M trainable parameters

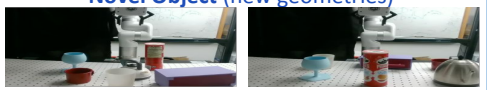
Appearance Shift (lighting, textures)



Scene A 27/30 | Scene B 23/30

0.493%
of full-model parameters
37M trainable parameters

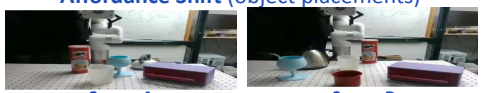
Novel Object (new geometries)



Scene A 26/30 | Scene B 21/30

0.800%
of full-model parameters
60M trainable parameters

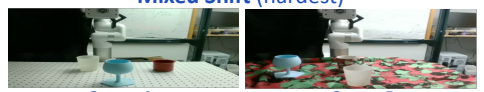
Affordance Shift (object placements)



Scene A 13/30 | Scene B 11/30

0.800%
of full-model parameters
60M trainable parameters

Mixed Shift (hardest)



Scene A 25/30 | Scene B 20/30

1.333%
of full-model parameters
100M trainable parameters